

Econ 101A

Section 1

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This first section reviews the mathematical toolkit from Lecture 1: optimization in one and several variables, the gradient and Hessian, and the Implicit Function Theorem. Almost every result in this course is an optimization problem in disguise, so it pays to be fluent with these tools.

A note on variables vs. parameters. We use a semicolon to separate the two. In $f(x_1, x_2; a)$, everything *before* the semicolon (x_1, x_2) is a **variable** you optimize over and take derivatives with respect to; everything *after* (a) is a **parameter**, a fixed constant.

1 Optimization with One Variable

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable. We look for an interior maximum (or minimum) x^* .

- **First-order condition (FOC):** $f'(x^*) = 0$. At an interior optimum the slope is flat.
- **Second-order condition (SOC):** $f''(x^*) \leq 0$ for a (local) maximum, $f''(x^*) \geq 0$ for a (local) minimum; strict inequality guarantees a strict local optimum.
- **Global optimality:** if f is *concave* everywhere ($f'' \leq 0$ on the whole domain), then any point satisfying the FOC is a *global* maximum. (Convex f : FOC point is a global minimum.)

Problem 1 (warm-up). A firm hires $L \geq 0$ units of labor to produce output $\sqrt{L}$, which sells at price $p > 0$; the wage is $w > 0$. Profit is $\pi(L) = p\sqrt{L} - wL$. Find the profit-maximizing L^* , check the SOC, and sign $\frac{dL^*}{dw}$.

2 A Little Multivariate Calculus

Consider a scalar function of many variables, $y = f(x_1, x_2, \dots, x_m) = f(\mathbf{x})$, where $\mathbf{x} = (x_1, \dots, x_m)'$ is a column vector.

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2.1 Gradient (first derivative)

The **gradient** of f is the column vector of partial derivatives:

$$\nabla f = \left[\frac{\partial f}{\partial x_i} \right] = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_m} \right]'$$

Problem 2. Derive the gradient of $y(x_1, x_2; a) = \ln(x_1) + e^{ax_2}$ over $(x_1, x_2) \in \mathbb{R}_{++} \times \mathbb{R}$.

2.2 Hessian (second derivative)

The **Hessian** H_f collects all second- and cross-partial derivatives:

$$H_f = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right] = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_m^2} \end{bmatrix}.$$

Provided f is smooth enough, H_f is symmetric (Clairaut/Schwarz: $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$).

Problem 3. Derive the Hessian of $y(x_1, x_2; a) = -(ax_1 - 1)^2 - (x_2 - 2)^2$.

3 Multivariate Optimization

Multivariate optimization is conceptually the same as the univariate case; we just need matrix versions of the FOC and SOC. For $y = f(\mathbf{x}; \boldsymbol{\alpha})$ with variables $\mathbf{x}$ and parameters $\boldsymbol{\alpha}$, an interior point $\mathbf{x}^*$ is a maximum (minimum) if:

- **FOC:** $\nabla f|_{\mathbf{x}=\mathbf{x}^*} = \mathbf{0}$.
- **SOC:** $H_f|_{\mathbf{x}=\mathbf{x}^*}$ is negative (positive) definite for a maximum (minimum).

The two-variable case (your main responsibility). A function $f(x_1, x_2)$ has a local maximum (minimum) at (x_1^*, x_2^*) if the FOC holds, $f'_{x_1} = f'_{x_2} = 0$, and

$$f''_{x_1 x_1} < (>) 0 \quad \text{and} \quad f''_{x_1 x_1} f''_{x_2 x_2} - (f''_{x_1 x_2})^2 > 0,$$

with all second derivatives evaluated at (x_1^*, x_2^*) .

Definiteness. For a symmetric $n \times n$ matrix H : H is *positive definite* if $\mathbf{z}' H \mathbf{z} > 0$ for every nonzero $\mathbf{z}$, and *negative definite* if $\mathbf{z}' H \mathbf{z} < 0$ for every nonzero $\mathbf{z}$ (replace $>, <$ by $\geq, \leq$ for semidefiniteness).

Sylvester's criterion (a handy check). Let $|H|_k$ be the k -th leading principal minor (the determinant of the top-left $k \times k$ block). Then:

- H is positive definite $\iff$ all $|H|_k > 0$;

- H is negative definite $\iff$ the signs alternate starting negative, i.e. $(-1)^k |H|_k > 0$ (so $|H|_1 < 0$, $|H|_2 > 0$, $|H|_3 < 0, \dots$).

A twice-differentiable f is concave (convex) iff its Hessian is negative (positive) semidefinite *everywhere*; then the FOC alone guarantees a global optimum.

4 Implicit Function Theorem (IFT)

What is it for? Doing comparative statics when you *cannot* solve the FOC in closed form. For example, when you want to know how optimal demand moves with price but cannot invert the first-order conditions.

Why does it work (intuition)? Suppose $f(x; p) \equiv 0$ defines x implicitly as a function of p . Totally differentiate with respect to p , remembering $x = x(p)$:

$$\frac{\partial f}{\partial p} + \frac{\partial f}{\partial x} \cdot \frac{dx}{dp} = 0 \implies \frac{dx}{dp} = -\frac{\partial f / \partial p}{\partial f / \partial x},$$

valid wherever $\partial f / \partial x \neq 0$. You know how f changes directly through p and indirectly through x ; since f stays equal to 0, you can back out how much p moved x .

5 Exercises

Exercise 1 (multivariate FOC/SOC; PS1, Spring 2014)

Let a and b be nonzero real parameters, and consider $\max_{x,y} f(x, y; a, b) = ax^2 - x + by^2 - y$.

- Write the first-order conditions with respect to x and y .
- Solve explicitly for the x^* and y^* satisfying the FOC.
- Using the second-order conditions at (x^*, y^*) , give sufficient conditions on a and b for this stationary point to be a maximum.
- Under what conditions on a, b is f concave in (x, y) ? When is it convex?

Exercise 2 (a saddle point)

Consider $f(x, y) = x^2 - y^2$ on $[-1, 1] \times [-1, 1]$.

- Find the maximum value of f and the points where it is attained (no calculus needed).
- Now apply the recipe: (i) write the FOC and solve for the stationary point; (ii) compute the Hessian there. Is it positive definite? Negative definite? What do you conclude?

Exercise 3 (Implicit Function Theorem)

The equation $f(x; p) = 1 - xp - p e^x = 0$ holds around $(x_0, p_0) = (0, 1)$. Use the IFT to find $\frac{dx}{dp}$ at that point.